

Pranav Doma

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EDUCATION:

New York University MS, Computer Engineering | CGPA -3.73 08/2022 - 05/2024
Coursework-Robot Perception, Computing Systems Architecture, Real-Time Embedded Systems, Deep Learning, Digital Signal Processing, Machine Learning, Applications of AI for Predictive Maintenance (NVIDIA DLI)
S. V National Institute of Technology B.Tech, Electrical Engineering | CGPA-7.24 07/2017 - 06/2021

SKILLS: C, C++, Embedded C, Python, CUDA, Matlab/Simulink, ROS, IceOryx/2, Git, Github Actions, Mujoco (MJX), Brax, OpenCV, PyTorch, ONNX, CARLA, Omniverse, IsaacSim/Lab, Motion Planning, Sensor Fusion, Visual SLAM, Embedded Systems for Robotics, Oracle Cloud, Linux, Docker, Podman, HPC (Slurm, Singularity), Atlassian, CrewAI, Groq, LangChain.

WORK EXPERIENCE:

HCLTech America- Robotics Application Engineer - (Autonomous Vehicles & Robotics) | New Brunswick 02/2026 - Present

- Led the transition from traditional modular pipelines to End-to-End (E2E) autonomous driving architectures, developing auto-labeling algorithms that utilize existing camera and radar perception models to generate supervision signals for training.
- Developed and released VisionPilot (all versions), an open-source Level 2 ADAS system for automotive OEMs, enabling real-time 3D world understanding on edge devices using a single front-facing camera. Built and integrated the longitudinal and lateral networks fusion and state-estimation modules using Kalman/particle filtering, and orchestrated system-level testing across SODA Sim, CARLA, and AlpaSim, with improved high-speed and sharp-turn driving performance.
- Built QA, inspection, and vision-based testing demos within HCLTech's Kinetic AI platform, including mobile phone QA testing with Vision-Language Models (VLMs), smart-glasses validation, visual inspection workflows, and robotic test automation using UR robots, LeRobot, Trossen WidowX AI, and Unitree Go2 for inspection.
- Developed multimodal robotic QA demonstrations leveraging ACT, VLA-JEPA, and NVIDIA GR00T 1.7, integrating RGB-D perception, multi-camera data collection, and manipulation workflows to showcase autonomous testing, inspection, and validation capabilities under the Kinetic QA framework.
- Built an end-to-end Physical AI pipeline on NVIDIA's open Alpa ecosystem, converting non-NVIDIA autonomous-driving data into simulation-ready environments via NCore/NuRec, integrating Fixer-enhanced novel-view rendering, and executing native Alpamayo closed-loop driving with reasoning-overlay visualization in AlpaSim.
- Developed a cross-platform AV demonstration framework combining reconstructed 3D scenes, VLM-based navigation, and Alpamayo trajectory planning on RTX PRO 6000 and NVIDIA Thor, delivering explainable driving behavior, closed-loop lane-change scenarios, and policy validation on custom data

MultiCoreWare Inc- Research Engineer (Embedded systems, Sensor Eng.) | Urbana Champaign 10/2024 - 02/2026

- Released VisionPilot v0.9 and v0.5, featuring a multi-threaded pipeline for sensor capture, inference, and logging/visualization in a Level-2 ADAS perception stack. Developed Closest In-Path Obstacle (CIPO) detection obstacle motion state estimation, and estimating safe distance using Mobileye RSS for AEB and longitudinal control, as well as a LaneTracker for maintaining cross-track error (CTE), yaw error, and road curvature estimates for lateral control leveraging a PID controller for reliable operation within the system's ODD.
- Developed a middleware-agnostic, portable model deployment framework compatible with ROS2, QNX, Zenoh, and Iceoryx, with CUDA, HIP, ONNX Runtime, and TensorRT for hardware-accelerated real-time inference, including data auditing and model quantization (QAT, PTQ).
- Performed real-world ADAS testing and validation with OEMs, including vehicle demos and performance evaluation.
- Developed empirical model for multi-sensor FMCW radar and lidar applications. Worked on Optimizing sparse MIMO antenna arrays using multi-objective desirability functions (DF), balancing PSLR and beamwidth (azimuth/elevation) for grid-based designs.
- Worked on a proof-of-concept for LLVM-MCToll targeting static binary translation between ARM and x86; focused on instruction-level debugging with GDB and validated translated binaries using Android Studio emulation.
- Device enumeration in the ACPI namespace, Developed Linux kernel modules; debugged I2C and UART communication.

New York University- Assistant Research Scientist | New York 05/2024 - 10/2024

- Developed a language-guided motion planning framework that achieved a 27.1% reduction in computational nodes and 39.1% improvement in instruction compliance over baseline planners.
- Designed a multi-agent LLM architecture for high-dynamic vehicle maneuvers, enabling iterative refinement of control parameters through physics-based validation, improving success rates to 90% through physics-based simulation validation.

TRC Robotics -Robotics Software Engineer & Business Development Associate | Remote 10/2023 - 10/2024

- Built and deployed Docker/Podman containers, implemented Grafana dashboards, and managed Keycloak authorization for ROS-based systems.
- Migrated servers and optimized CI/CD pipelines, enhancing system reliability and deployment efficiency.

PROJECTS:

RL-Based Quadruped and Humanoid Control Learning (Pytorch, MJX, Brax, Tensorflow)

- Developed and optimized PPO reinforcement learning pipelines using Brax/JAX for simulated quadruped and humanoid locomotion within the MJX physics engine.
- Engineered and debugged robust checkpointing systems inside Brax training loops, ensuring reliable model saving and resumption. Scripted end-to-end RL workflows integrating environment setup, parallelized training, policy evaluation, and results visualization (video/HTML outputs).

CUDA GPU Programming Portfolio | (CUDA, Cpp, SIMD)

- Implemented a suite of CUDA applications (vector addition, matrix operations, convolution, softmax, image processing) with memory coalescing and shared memory tuning.
- Built SIMD-optimized CPU versions using AVX2/SSE instructions for direct performance comparison.
- Built custom CUDA conv2D kernels on image data sourced from OpenCV.

Visual Navigation Game & 3D Mapping | (OpenCV, Sensor Integration, Python, Pygame, ROS)

- Engineered a monocular first-person view solution with SIFT for feature extraction; enabled maze exploration and image target navigation.
- Adapted the system for real-world TurtleBot3 deployment; utilized IMU and camera sensors for localization and mapping.
- Data collection for 3D construction mapping: 360° cameras, 3D LiDARs, IMU, LattePanda SBC, DeepMapping2.

PUBLICATIONS:

- **Geometry-First Generative Worlds as Simulation Substrates in Bronchoscopy -SurgTwin @ IROS 2026**
- **Immersive Social Interaction with VR and LLM-Assisted Humanoids -arXiv**
- **Maneuver GPT-Agentive Control for Safe Autonomous Stunt Maneuvers - IROS 2025**
- **LLM-Enhanced Path Planning: Safe and Efficient Autonomous Navigation with Instructional Inputs-arXiv**
- **Explainable Autonomous Mobile Robots: Interface and Socially Aware Learning -Late Breaking Report, IEEE-ROMAN-2024**
- **A Review On Wind Farm Reliability With Hybrid Cable Connection** - Book Chapter, Energy Conversion Systems: An Overview, Nova Science Publishers. ISBN: 978-1-53619-200-1